

Quiz (Carolina Reaper)

- Q1.** A truck travels $2x+5$ miles in x hours. If it travels 17 miles, how many hours did it take?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** Solve for x : $x - 3 = 7$
(A) 4
(B) 5
(C) 10
(D) 11
(E) 12
- Q3.** A shipping company charges 5 dollars per package plus an additional $2x$ dollars for delivery, where x is the number of miles traveled. If the total cost is 17, how many miles were traveled?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** Solve for y : $y + 2 = 11$
(A) 8
(B) 9
(C) 10
(D) 12
(E) 13
- Q5.** A traffic flow model predicts that the number of cars on the road will be $3x^2 + 2x - 5$ after x hours. If there are 10 cars on the road, how many hours have passed?
(A) 1
(B) 2
(C) -1
(D) 3
(E) 4
- Q6.** Solve for z : $z - 2 = 9$
(A) 7
(B) 10
(C) 11
(D) 12
(E) 13
- Q7.** A delivery driver must travel $x + 10$ miles to reach the destination. If the driver has traveled $x - 5$ miles so far, how many miles are left to travel?
(A) 15
(B) 10
(C) 5
(D) 20
(E) 25
- Q8.** Solve for w : $2w + 5 = 11$
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Open Ended Questions (FRQ)

- Q9.** A package is traveling from City A to City B at a rate of x miles per hour. If the distance between the two cities is $3x + 10$ miles and the package takes $x + 2$ hours to arrive, solve for x and check your solution for accuracy. Show all work.
- Q10.** A logistics company has $5x + 2$ packages to deliver in x hours. If each package requires $2x - 1$ minutes to deliver, how many packages can be delivered per hour? Solve for x and check your solution for accuracy. Show all work.

Chef's Tips

- Tip 1: Encourage students to check their solutions for accuracy
- Tip 2: Remind students to use the correct order of operations when solving equations
- Tip 3: Suggest that students use real-world examples to illustrate their solutions